COMP1850 Programming

School of Computer Science, University of Leeds

Worksheet 2.3

This worksheet contains a combination of formative activities and activities that contribute towards your portfolio.

- Portfolio activities are indicated in the title of the task.
- Activities marked by (*) are advanced activities and may take more time to complete.

Expectations:

1. Timeliness: You should complete all the tasks in the order provided and submit your portfolio task to Gradescope before the deadline.
2. Presentation: You should present all your work clearly and concisely following any additional guidance provided by the module staff in the module handbook.
3. Integrity: You are responsible that the evidence you submit as part of your portfolio evidence is **entirely your own work**. You can find out more about Academic integrity on the Skills@library website. All work you submit for assessment is subject to the academic integrity policy.

Feedback: Feedback on formative activities will be provided via Lab classes and tutorials. Feedback on evidence submitted as part of the portfolio will be available on Gradescope.
Support opportunities: Support with the activity sheet is available in the Lab classes and tutorials. Individual support is available via office hours.
Expected complete date: Friday 20th February

All portfolio tasks are rated AMBER by the university: AI tools can be used in an assistive role. You are permitted to use AI tools for specific defined processes within the assessment.

Within this assessment you **may** use Generative AI to:

- help you start to understand more complex ideas by providing accessible summaries
- test your knowledge against a piece of content
- help identify and correct spelling mistakes and grammatical errors in your work
- provide feedback and advice on your overall coding style

You must **not** use Gen AI to:

- produce written content (code, comments or any other content) which you then submit as your work, regardless of whether you make changes to it.
- rewrite or make changes to any of your work.

Learning Outcomes:

- Explain and understand the differences between compiled and interpreted languages, including the benefits and limitations of each.
- Develop programs that use the fundamental programming constructs: assignment and expressions, basic I/O, conditional and iterative statements.
- Use appropriate terminology to identify elements of a program (e.g., identifier, operator, operand).

Practice tasks

Implement and test C programs for the following tasks.

In each case consider which data type you should use and how to combine it with other data.

A brief description of the task is given inside each code template:

- var_test.c
- combine_vars.c
- pointer_test.c
- addresses.c

Portfolio Task

There are 2 tasks that require completing for the portfolio:

- Coding task – submission via Gradescope
- MCQ – submission on Minerva

You must pass all tests and all MCQ questions to complete the worksheet.

- Test your code in the terminal before submitting to Gradescope.
- Some MCQ questions refer to the coding task.

Description:

You are provided with the following data:
- Salary in £ (UK pounds), which is in the range of £14000-£40000
- National Insurance (NI) rate as a %, which is in the range 0%-10%
- Tax rate as a %, which is in the range 10%-30%

You should calculate the take-home salary after deductions according to the following rules:
- National Insurance is deducted from the total salary
- Tax is deducted from the salary remaining after NI deduction, and only applies to the part of salary remaining over £12500

Your program should print:
- The NI contribution
- The tax contribution
- The take home salary

in an appropriate way.
Eg. It is usual to write an amount of money with 2 decimal places.
In C you can modify the float format `%f` to print to 2dp with `%.2f`

As part of your implementation you should decide the appropriate data types for your variables and the appropriate form of the expressions.

It is important to test your solution. One way to consider this is validate with simple test cases.
- NI=0%, tax=0% should produce no deduction
- NI=0%, tax=100% should result in take home salary £12500 in every case
- NI=10%, tax=0% should simply reduce salary by 10%

To submit your code you should set salary to £36250, NI rate to 8%, tax rate to 15%.

Use only the print statements provided to output data.

Your code should use sensible naming of variables and use appropriate whitespace and inline comments.

Sample Exam Question

State a different feature for each language due to the fact that:

- C has pointers but Python does not
- C has primitive data types but Python does not

Describe how pointers in C are used to implement the handling of an integer variable in the Python language.